# Rejection Sampling for Medical MCQ Generation and Fine-Tuning TinyLlaMA for Layperson QA

AI in Healthcare
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# Introduction

# Background

- Medical QA: complex, needs accuracy
- Layperson QA: needs simplicity
- LLMs: need domain fine-tuning
- Good data is critical

# Objectives

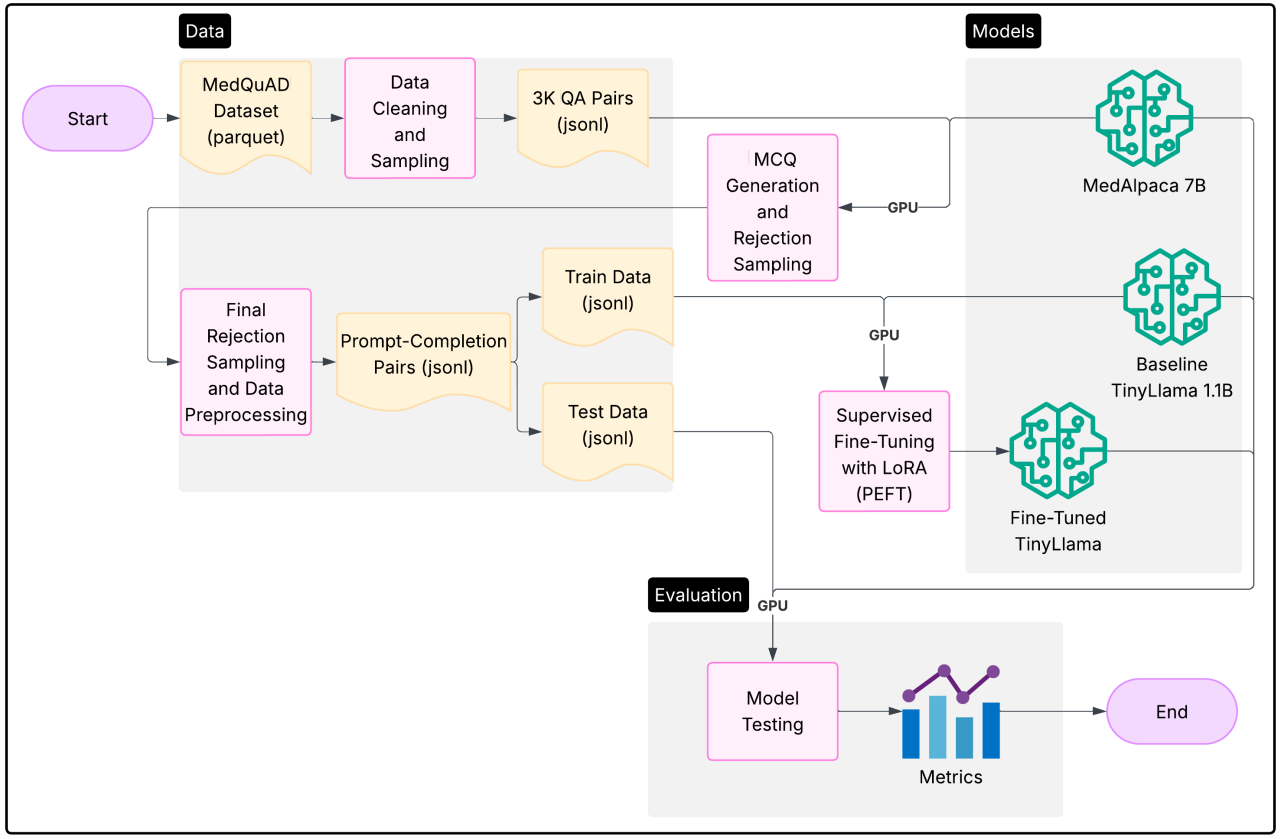
- Generate MCQs from MedQuAD
- Filter with rejection sampling
- Fine-tune TinyLLaMA
- Compare to baseline and MedAlpaca

# Method

# MedQuAD Dataset

- 47K+ layperson medical QA pairs
- Trusted U.S. health sources
- Clear, accessible answers

# Workflow


Data
Models
Start
MedQuAD Dataset (parquet)
Data Cleaning and Sampling
3K QA Pairs (jsonl)
MCQ Generation and Rejection Sampling
GPU
MedAlpaca 7B
Final Rejection Sampling and Data Preprocessing
Prompt-Completion Pairs (jsonl)
Train Data (jsonl)
Test Data (jsonl)
GPU
Baseline TinyLlama 1.1B
Supervised Fine-Tuning with LoRA (PEFT)
Fine-Tuned TinyLlama
Evaluation
GPU
Model Testing
Metrics
End

# MCQ Generation

```
# Few-shot example to encourage structure
few_shot_example = """
Question: What is asthma?
Answer: Asthma is a lung condition that causes airway inflammation and breathing difficulty.
<choices>
A. A skin condition
B. A viral infection
C. A lung disease that causes airway inflammation
D. A type of heart failure
</choices>
<reason>Asthma is a lung condition that causes airway inflammation and breathing difficulty.</reason>
<answer>C</answer>
"""

# Format QA into prompt
def format_prompt(question, answer):
    return f"""
Create layperson multiple-choice answer choices and reasoning for the given medical Q&A pair.
Include 4 total choices: one correct and three plausible, but incorrect. Follow the XML-style structure below:
{few_shot_example}
Now it's your turn:
Question: {question}
Answer: {answer}
"""
```

# Generated MCQs


```
{"prompt":"You are a helpful and concise medical assistant.\n\nAnswer the
following multiple-choice question. First, briefly explain your reasoning using
relevant medical knowledge. Then state your final answer letter using exactly one
<answer></answer> tag.\n\nInstructions:\n- Use only one <answer> tag (e.g.,
<answer>B</answer>)\n- Choose the best answer letter\n- Stop immediately after
the </answer> tag\n\n<question>\nWhat causes Gum (Periodontal) Disease?\nA.
Genetic mutations\nB. Poor dental hygiene\nC. Injury to the gums\nD. Periodontal
abscesses\n</question>\n","completion":"Gum disease is caused by dental plaque --
a sticky film of bacteria that builds up on teeth. Regular brushing and flossing
help get rid of plaque. But plaque that is not removed can harden and form tartar
that brushing doesn't clean. Only a dentist or dental hygienist can remove
tartar.\n<answer>B</answer>"}
```


- 511 MCQs finalized
- 408 train / 103 test (80/20 split)

# Results

# Loss

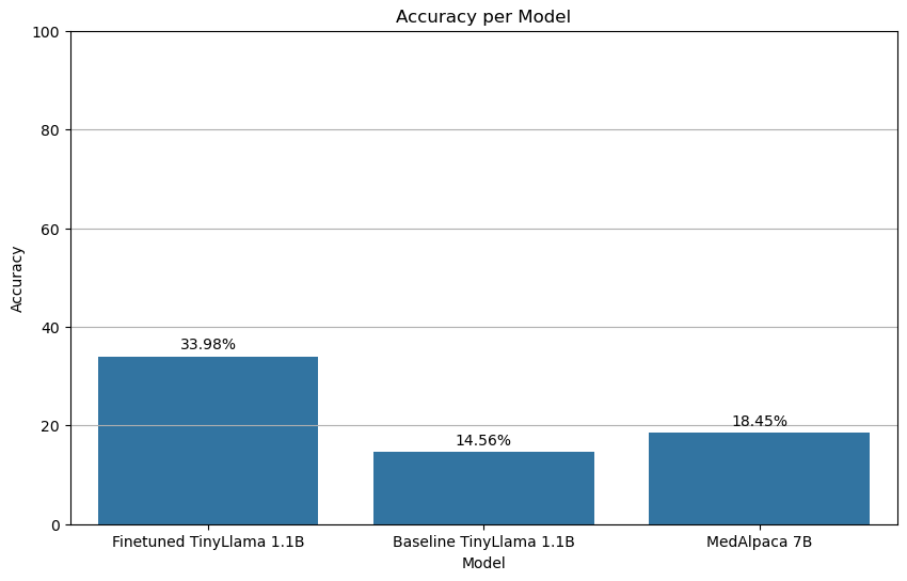
Small dataset: quick, noisy training

[48/48 06:11, Epoch 3/3]

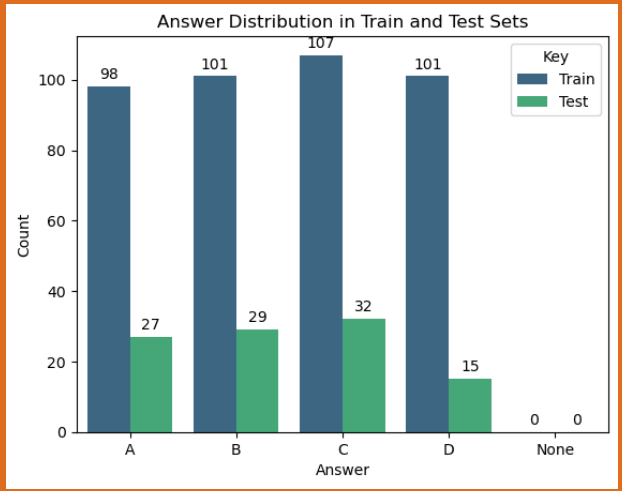
| Epoch | Training Loss | Validation Loss |
|---|---|---|
| 1 | 0.846100 | 0.489880 |
| 2 | 0.370500 | 0.475589 |
| 3 | 0.363300 | 0.481885 |

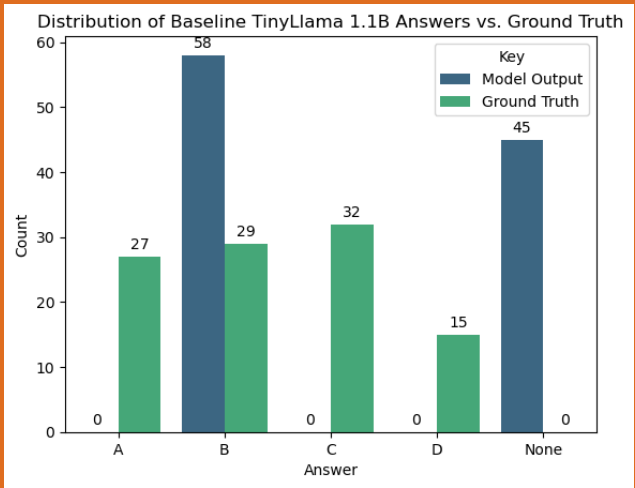
# Accuracy

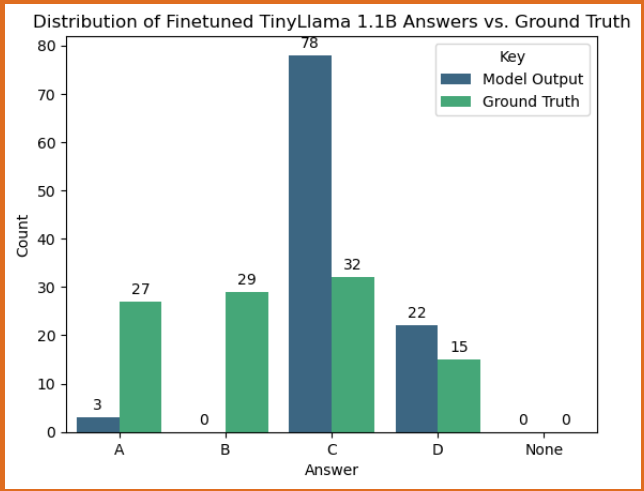
Finetuning improved accuracy, but gains were modest

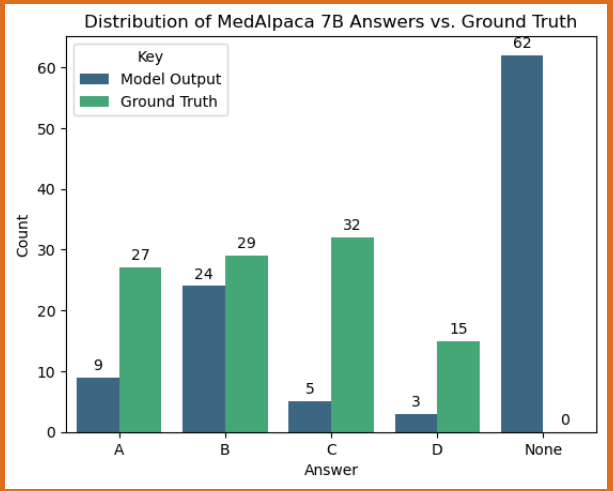


# Answer Distributions



# Future Directions

- Scale to larger models
- Improve MCQ generation
- Extend to open-ended QA
- Test on broader datasets

# Thank You!